TIGA WARNA PRIMER

TEXTILE DYE AND CHEMICAL SPECIALIST

Disperse Dye

ENGINEERED FOR BRILLIANT COLOR

HIGH-PERFORMANCE DYES
& CHEMICALS SOLUTIONS
FOR TEXTILE EXCELLENT

INNOVATIVE CHEMISTRY

VIBRANT COLOR

SUSTAINABLE FUTURE

DISPERSE

About Us

PT Tiga Warna Primer is a manufacturer of textile dyes and specialty chemicals. We are committed to delivering innovation, superior product quality, and professional technical support, with customer satisfaction at the heart of everything we do. By providing efficient, reliable, and environmentally responsible solutions, we help our customers achieve outstanding performance while supporting a more sustainable textile industry.

TIGA WARNA PRIMER

ZDHC

OEKO TEX® STANDARD 100

CERTIFIED ISO 9001 CERTIFIED

BANGGA BUATAN INDONESIA

DISPERSE

TIGA WARNA PRIMER

Dyeing Process Using Dispersion Dyes

I. High-Temperature and High-Pressure Dyeing

1. This method uses a closed-loop dyeing machine. Depending on the form of the fabric to be dyed, various dyeing methods are used, such as twisted yarn dyeing, bobbin yarn dyeing, yarn-on-warp dyeing, bobbin dyeing, fluid-flow dyeing, and spray dyeing.
2. Generally, 1–2 g/L of dye dispersion is added to the dye solution, using a buffer such as acetic acid or sodium acetate to adjust the pH of the dye solution to 5–6. The dyeing process then begins at a temperature of 40–60°C, which is gradually raised to 125°C–135°C, and continues for 30–90 minutes.
3. During the dyeing process, light colors do not always require the addition of a dispersant, whereas for medium to dark colors, a dispersant must be added to maintain the dispersion of the dye solution.
4. For some types of disperse dyes, a dye solution with a pH that tends to be alkaline can affect the degree of color absorption. To achieve good dyeing results, the pH of the dye solution must be adjusted, while the dyeing temperature and time depend on the type of material being dyed, the desired depth of color, and the dyeing equipment, so they must be selected appropriately.
5. After the dyeing process is complete and the fabric has been rinsed with water, perform a reduction wash or a soap wash to remove any floating dye residues and improve colorfastness to rubbing. Conditions: 38°Be, 3 mL/L NaOH, 2 g/L sodium amidosulfate (or 0.5 g/L reducing agent), temperature 80°C × (10–15)

II. Dyeing Method with a Carrier

1. This method involves adding a carrier to the dyeing solution, followed by dyeing at normal pressure. Compounds such as methyl naphthalene and chlorobenzene are used as carriers. For various types of carriers, the amount added to the dyeing solution is generally 2–7 g/L; other dyeing conditions are the same as for the high-temperature, high-pressure dyeing method, namely starting the dyeing at 40–60°C, slowly raising the temperature to 100°C, and continuing the dyeing for 90–120 minutes.
2. In carrier-based dyeing, because the carrier is emulsified using an emulsifier, dyeing defects such as color differences between the beginning and end of the fabric or carrier spots may sometimes occur; therefore, it is very important to select a carrier that remains stable throughout the dyeing process.
3. The reduction wash is the same as in the high-temperature, high-pressure dyeing method.

III. Hot-Melt Dyeing Method

1. This continuous dyeing method is often used to dye blended fabrics such as polyester/cotton.
2. In addition to disperse dyes, the dyeing solution typically contains 0.5–2 g/L of an anti-migration agent such as sodium alginate, as well as a specific amount of a penetration agent if necessary.
3. The fabric is immersed in a dispersion dye solution to a 65–70% absorption level; after intermediate drying, the fabric undergoes a hot-melt process. The hot-melt conditions depend on the properties of the dispersion dye and
4. The type of fabric being dyed, typically range from (180–210 °C) × 60–120 seconds.
   Recommendations: Type E 180–190°C × 60–120 seconds,
   Type SE 190–200°C × 60–120 seconds,
   Type S 200–210°C × 60–120 seconds
5. Reduction wash, same as in the high-temperature, high-pressure method

IV. Printing

Classification based on printing and dyeing methods: 1. Direct printing
2. Printing using a color-removal technique
3. Printing using anti-dyeing techniques

Based on printing equipment:
1. Manual printing
2. Screen printing
3. Copper cylinder printing

1. Direct printing

(1) High-Pressure Evaporation (HPS) Method

Method Color paste formulation	
Dispersion dyes	x
Warm water (50–60 °C)	y g
Raw dough	500–600 g
Tartaric acid	3 g
Sodium chlorate	5 g
Total amount	1000 g

(2) High-Temperature Vaporization (HTS)

Dispersion Pigments	x g
Warm water (50–60 °C)	y g
Original dough	500–600 g
Color enhancer	20–40 g
Tartaric acid	3 g
Sodium chlorate	5 g
Total	1000 g

Molding → Drying → Saturated steam evaporation (130°C × 30`) → Water washing to remove residual binder → Hot water washing → Reduction washing → Water washing to remove residual preservative → Hot water washing → Reduction washing

Molding → Drying → Hot steam evaporation (180–185°C × 6–8`)

2. Printing using the color-removal technique

Dispersion dyes that can be removed are used for dyeing via the carrier method or the high-temperature, high-pressure method, followed by color removal using the following method. Lead chloride is used as the color-removing agent.

(1) Printing with white color removal

Lead(II) chloride	80–120 g
Tartaric acid or oxalic acid	15–25 g
Urea	25 g
Water	115–165 g
Base Paste	750–665 g
DT bleaching dye	15 g

(2) Printing using white color removal

Dispersion dye (lead chloride)	x g
Lead(II) chloride	50–80 g
Urea	80 g
Dyeing auxiliary (hydroquinone)	50 g
Water	x g
	800–700 g

Printing → Drying → Evaporation at 120–130°C × 20–30` → Water wash to remove paste → Hot water wash followed by a reduction wash.

3. Anti-stain printing

Anti-dye paste recipe		Anti-stain paste recipe for overprinting Colorfast dispersion	
dyes	x g	Wash-out dispersion dyes	x g
Original paste	500–600 g	Raw paste	500–600 g
Penetration	0–20 g	Penetration	0–20 g
agent	y g	agent	y g
Warm		Warm	
water		water	
Lead(II) chloride	40–60 g	Tartaric acid	3–5 g
Tartaric acid	3–5 g	Sodium chlorate	2 g
Total	1000 g	Total	1000g

Apply the anti-stain paste → Allow to dry or print immediately → Apply the next layer of anti-stain paste → Steam at 120–130°C for 20–30` → Rinse with water to remove the paste → Rinse with hot water → Rinse to remove residual paste.

V. Durability Test

Lightfastness (xenon lamp)	GB/T 8427-2008 (setara dengan ISO 105-B02:1994)
Washfastness	GB/T 3921-2008 C(3) (setara dengan ISO 105-C10-2006)
Color Fastness to Rubbing	GB/T 3920-2008 (setara dengan ISO 105-X12-2001)
Color fastness to sublimation	GB/T5718-1997 (setara dengan ISO 105P01:1993)

VI. Notes: ○ Most suitable ◎ Suitable △ Not suitable

Starcron Dye Dyeing Procedure

1. High-temperature and high-pressure dyeing:

1.1. High-temperature, high-pressure dyeing is performed using closed-type dyeing machines such as the Chease, Beam, JAT, Jig, or solution-flow types, depending on the shape of the fabric.
1.2. General dyeing conditions are as follows:
1.3. Approximately 1–2 g/L of a dispersing agent is added to the dyeing solution, and the solution's pH is then adjusted to 5–6 using acetic acid or a buffer solution (acetic acid and sodium acetate). The dyeing bath is then gradually heated from 40–60 °C to 125–135 °C, and at this temperature, the dyeing process continues for 30–90 minutes.
1.4. Furthermore, the dyeing temperature and time vary depending on the type of material, dye concentration, and dyeing equipment. Therefore, it is important to select appropriate dyeing conditions.
1.5. After the dyeing process using various methods, a reduction wash is generally performed; dyes that have not yet bound to the fiber surface are removed through this treatment.
1.6. Good color fastness, as well as good rub fastness and wet fastness, can be achieved.
1.7. The conditions for the reduction wash are generally as follows: 2 g/L hydrosulfite
3 ml/L caustic soda at 38°Be Temperature:
80°C Time: 10–15 minutes

2. Dyeing with a carrier

2.1. Dyeing with a carrier is performed at atmospheric pressure by adding the carrier to the dye solution. Carriers such as methylnaphthalene or chlorobenzene are commonly used, and are added at a concentration of 2–7 g/L to the dye solution. The other conditions of the dye solution are the same as those used in high-temperature dyeing. The temperature is gradually raised from 40–60°C and maintained for 90–120 minutes at 100°C.

2.2. In carrier dyeing, in some cases, problems such as dye precipitation or carrier staining are caused by the instability of the emulsifier in the carrier at high temperatures or under other conditions. Therefore, it is important to select the appropriate carrier to prevent such problems.

2.3. The reduction wash is performed as described for high-temperature, high-pressure dyeing.

3. Thermosol Dyeing

3.1. Thermosol dyeing is a continuous dyeing method primarily applied to polyester/cotton blends. The typical dyeing conditions are as follows.

3.2. Dye and 0.5–2 g/L of a migration inhibitor (e.g., sodium alginate), as well as a wetting agent if necessary, are added to the padding solution.

3.3. Squeezing is performed using the “one dip, one squeeze” method at room temperature, with an absorption rate of 65–70%. Drying is carried out at 70–90°C.

3.4. After the padding and intermediate drying processes, thermosol fixation is performed by dry heating for 60–120 seconds at 180–210°C, depending on the fabric type and the type of Starcron dye.

Type E: 180–190°C × 60–120 seconds.

Type SE: 190–200°C × 60–120 seconds.

Type S: 200–210°C × 60–120 seconds.

Reduction cleaning as described in the high-temperature, high-pressure dyeing process.

4. Printing

Printing methods are generally classified as follows.

Based on printing style	Direct printing
	Discharge printing
	Resist printing
	Hand printing
Based on printing tools	Screen printing
	Roller printing
	Others

4.1 Direct printing

(1.) High-pressure steaming (HP steaming)

Color paste recipe:

Starcron Pigment	xg
Water (50–60 °C)	yg
Thickening paste	500–600
g Tartaric acid	3
g	
Sodium chlorate	5 g
Total	1000 g

After molding and drying, steam for 30 minutes at 130 ℃, then wash with cold and warm water, and perform a reduction wash.

(2.) High-Temperature Steaming (HT Steaming) Method Starcron Dye Xg Water (50–60 °C) Yg

Thickening paste	500–600 g
Accelerate coloring	20–40 g
Tartaric acid	3 g
Sodium chlorate	5 g
Total	1000 g

After printing and drying, steam for 20–30 minutes at 120–130 ℃, then wash with cold and warm water, followed by a reduction wash.

4.2 Discharge printing

Fabrics dyed with dischargeable Starcrons dyes using the carrier method or high-temperature immersion are decolored as follows: Using stannous chloride as the discharge agent.

(1) White discharge

Stannous chloride	80–120 g
Tartaric acid	15–25 g
Urea	25 g
Water	115–165 g
Thickening paste	750–665 g
Fluorescent brightening agent DT	15 g
Total	1000 g

(2) Colored discharge dyes

Starcron non-removable dye	Xg
Stannous chloride	50–80 g
Urea	50 g
Auxiliary materials for the release process	50 g
Water	yg
Total	1000 g

After printing and drying, steam for 20–30 minutes at 120–130 ℃, then wash with cold and warm water, followed by reduction cleaning.

4.3 Resist Printing

Resist paste recipe:

Starcron non-removable dye Xg

Thickening paste	500–600 g
Wetting agent	0–20 g
Warm water	that
Stannous chloride	40–60 g
Tartaric acid	3–5 g
Total	1000 g

Pasta Recipe:

Removable Starcron coloring	Xy
Thickening Paste	500–600 g
Warm water	yg
Tartaric acid	3–5 g
Sodium chlorate	2 g
Total	1000 g

After printing the resist paste, proceed with printing the top layer—either immediately or after drying—then dry it, steam it for 20–30 minutes at 120–130 ℃, wash it with cold and warm water, and finally perform a reduction wash.

5. Ketahanan

Light resistance (xenon)	GB/T 8427-2008 (setara dengan ISO 105-B02-1994)
Resistance	GB/T3921-2008 C(3) (setara dengan ISO 105-C10-2006)
Abrasion Resistance	GB/T3920-2008 (setara dengan ISO 105-X12-2001)
Resistance to sublimation	GB/T5718-1997 (setara dengan ISO 105P01:1993)

6. Notes

◯ Highly suitable
◎ Suitable
△ Not suitable

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
1.		Yellow TSE-FL 42#	1.0	○	◎	◎	7	3-4	4-5	4-9
2.		Yellow TE-3G 54#	1.0	○	◎	◎	6	4	4-5	4-8
3.		Yellow TSE-4RL 56#	1.0	○	◎	◎	6	4-5	4-5	4-8
4.		Yellow TSE-3GL 64#	1.0	○	◎	△	7	4	4	3-9
5.		Yellow THL-5G 71#	2.0	○	△	◎	7	3-4	4	4-6
6.		Yellow TS-G 79#	1.0	○	◎	◎	6	4-5	4-5	4-8
7.		Fluorescent Yellow T-8GFF 82#	1.0	○	○	◎	4	4-5	4-5	4-6
8.		Yellow TS-6GSL 114#	1.0	○	○	○	7	5	5	4-6
9.		Yellow THW-GL 118#	1.0	○	○	◎	6-7	4-5	5	4-9
10.		Yellow TSE-5GL 119#	1.0	○	○	◎	6-7	4-5	4-5	4-6
11.		Yellow THL-2GL 122#	1.0	○	○	△	7	5	4-5	4-9

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
12.		Yellow THW-7G 126#	1.0	○	○	◎	6-7	5	4-5	4-9
13.		Yellow TS-BRL 163#	2.0	○	○	○	7	5	4-5	4-8
14.		Fluorescent Yellow T-10GN 184:1#	1.0	○	△	◎	5	4	4	3-6
15.		Yellow TSE-4G 211#	1.0	○	◎	◎	6	3-4	4	4-6
16.		Yellow TSE-5GR 241#	1.0	○	○	◎	6-7	4-5	4-5	3-9
17.		Yellow THW-3R 246#	1.0	○	○	○	6-7	5	4-5	4-9
18.		Golden Yellow TSE-3R MIX	1.0	○	△	◎	6	4-5	4	3-6
19.		Orange TE-RL 25#	2.0	○	△	◎	5-6	3-4	4	3-9
20.		Orange TSE-3GL 29#	2.0	○	◎	◎	6	4-5	4-5	4-8
21.		Orange TS-4RL 30#	2.0	○	○	○	6	4-5	4-5	4-7
22.		Orange TSE-2R 31#	2.0	○	○	◎	6	4-5	4	4-7

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
23.		Orange TS-3RFL 44#	1.0	○	○	○	6	4-5	4	4-8
24.		Orange TS-2RL 61#	2.0	○	◎	◎	6	4	4-5	3-9
25.		Orange TSE-3RL 62#	2.0	○	○	○	6-7	5	4-5	4-6
26.		Orange TS-SF 73#	2.0	○	○	◎	5	5	4-5	3-9
27.		Orange TSE-3RLS 80#	1.0	○	◎	○	5-6	4-5	4-5	4-6
28.		Orange TSE-RFL 288#	2.0	○	◎	◎	5-6	4	4-5	3-9
29.		Fluorescent Orange T-2GFL	2.0	○	△	◎	4	4	4	4-6
30.		Red TT-B 1#	1.0	○	◎	◎	5-6	3-5	4	4-7
31.		Red TT-BD 13#	1.0	○	◎	◎	4-5	3-4	4-5	4-6
32.		Red TT-2GH 50#	1.0	○	○	◎	5	3-4	4-5	3-9
33.		Red TS-3GL 54#	2.0	○	○	◎	6-7	4	4-5	4-7

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
34.		Red TE-FB 60#	2.0	○	◎	◎	6-7	3-4	4-5	3-9
35.		Red TSE-GFL 73#	1.0	○	○	◎	6	3-4	4-5	4-7
36.		Red TS-BGL 74#	2.0	○	○	◎	5-6	3-4	4	4-6
37.		Red TSE-BBL 82#	2.0	○	○	◎	6-7	4	4	4-6
38.		Red THL-5GL 86#	2.0	○	○	◎	6-7	5	4-5	4-6
39.		Red THL-REL 91#	2.0	○	○	◎	6-7	5	4	4-6
40.		Red TS-BEL 92#	2.0	○	○	◎	6-7	4-5	4	3-9
41.		Red TT-2BL 145#	1.0	○	○	◎	6-7	4	4	4-6
42.		Red TT-BL 146#	2.0	○	◎	◎	6	4	4	3-9
43.		Red TT-BS 152#	1.0	○	○	◎	5	4	4-5	4-8
44.		Red TT-GS 153#	1.0	○	○	◎	6	3-4	4-5	4-8

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
45.		Red TSE-4GFL 153:1#	1.0	○	○	◎	6	4-5	4-5	4-8
46.		Red TS-2GFL 167#	2.0	○	○	○	6-7	4	4-5	4-6
47.		Red TT-FRL 177#	1.0	○	○	○	5-6	5	4-5	4-8
48.		Red TT-2B 179#	1.0	○	○	◎	5-6	5	4-5	4-8
49.		Fluorescent Red T-G(Y) 277#	1.0	○	○	◎	5	4-5	4-5	4-8
50.		Fluorescent Red T-G(B) 277#	1.0	○	○	◎	5	4-5	4-5	4-8
51.		Red THS-XF 311#	2.0	○	◎	◎	5-6	4-5	4-5	4-9
52.		Red TSE-3B 343#	2.0	○	○	◎	5-6	4-5	4-5	3-9
53.		Red TSE-3BN 343:1#	2.0	○	◎	◎	5-6	4-5	4-5	4-9
54.		Red THW-CBN 356#	2.0	○	○	◎	6	5	4-5	4-8
55.		Red THW-BWF 356 : 2#	2.0	○	○	◎	6-7	4-5	4-5	4-8

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
56.		Red T-FBS 362#	2.0	○	◎	△	5-6	4	4-5	4-6
57.		Fluorescent Red T-BG(Y) 362:1#	1.0	○	◎	△	6	4	4-5	4-6
58.		Red THS-SF 367#	2.0	○	◎	◎	6	4-5	4-5	3-9
59.		Red THW-4BSN 367:1#	2.0	○	○	◎	6-7	5	4-5	4-8
60.		Fluorescent Red T-GG	2.0	○	○	◎	4-5	4	4	4-6
61.		Red TE-3B	2.0	○	○	◎	5-6	4-5	4-5	3-9
62.		Violet T-2R 1#	2.0	○	◎	△	6	4	4	4-8
63.		Violet TT-FBL 26#	2.0	○	○	○	6-7	3-4	4-5	4-6
64.		Violet THL-2RL 28#	2.0	○	△	◎	6-7	3-4	4-5	4-6
65.		Violet TT-CB 33#	2.0	○	○	◎	6	3-4	4-5	4-6
66.		Violet TT-3RL 63#	2.0	○	○	◎	6-7	4-5	4-5	4-6

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
67.		Violet TT-5R 93#	1.0	○	○	◎	6-7	3-4	4-5	4-8
68.		Blue TT-FFR 3#	2.0	○	◎	◎	6	4	4-5	4-8
69.		Brilliant Blue TE-2BLN 56#	2.0	○	◎	◎	6-7	4-5	4-5	3-9
70.		Brilliant Blue TE-4R 56#	2.0	○	◎	◎	7	3-4	4-5	3-9
71.		Turquoise Blue TS-GL 60#	2.0	○	○	○	7	5	4-5	3-9
72.		Blue THL-BGL 73#	2.0	○	○	◎	7	5	4	3-9
73.		Blue THL-BFL 77#	2.0	○	○	◎	6-7	4-5	4-5	4-8
74.		Blue TS-HGL 79#	2.0	○	○	○	5-6	4-5	4-5	4-6
75.		Brilliant Blue THL-BGF 87#	2.0	○	○	◎	6	5	4-5	4-6
76.		Blue THA-GF 102#	2.0	○	○	◎	6-7	5	4-5	4-8
77.		Blue TT-3RT 148#	2.0	○	○	◎	6	4-5	4-5	4-6

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				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
78.		Blue TT-BBLS 165#	2.0	○	○	◎	5-6	4-5	4	4-6
79.		Blue TT-BGLS 165 : 1#	2.0	○	○	◎	5-6	4-5	4	4-6
80.		Blue TSE-2R 183#	1.0	○	○	◎	5-6	4	4-5	4-6
81.		Blue THS-3RF 257#	2.0	○	△	△	6	4-5	4-5	4-7
82.		Blue THS-AR 281#	2.0	○	○	◎	5-6	4-5	4-5	4-6
83.		Blue THS-XF 284#	2.0	○	◎	◎	5-6	4-5	4-5	4-8
84.		Blue TT-3G 291#	2.0	○	○	◎	6-7	4	4-5	4-6
85.		Blue TT-3GR 291:1#	2.0	○	○	◎	6-7	4	4-5	4-8
86.		Blue TS-3G 301#	1.0	○	○	◎	5-6	4-5	4-5	4-6
87.		Blue TS-R 354#	2.0	○	○	△	5-6	4-5	4-5	4-6
88.		Blue TT-CRE 366#	1.0	○	◎	◎	6	3-4	4-5	4-6

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
89.		Blue THS-2GS 367#	2.0	○	◎	△	5-6	4	4	4-9
90.		Blue THS-2GSN 367:1#	2.0	○	△	△	5-6	4	4-5	4-9
91.		Blue TT-2BLM MIX	2.0	○	◎	◎	6	3-4	4-5	4-6
92.		Deep Blue TT-SF MIX	2.0	○	◎	◎	6	3-4	4-5	4-6
93.		Blue TT-3RN	2.0	○	◎	◎	6	3-4	4-5	4-6
94.		Deep Blue T-ECO MIX	2.0	○	△	◎	6	3-4	4-5	4-8
95.		Green TT-6B 9#	2.0	○	◎	◎	5-6	4	4-5	3-7
96.		Fluorescent Apple Green T-5G	2.0	○	◎	◎	5	4-5	4-5	4-6
97.		Brown THS-XF 19#	2.0	○	◎	◎	5-6	4-5	4-5	4-6
98.		Brown TPP-2BL MIX	2.0	○	○	○	5	4-5	4-5	4-6
99.		Grey T-N MIX	1.0	○	◎	◎	5-6	4-5	4-5	4-8

No.	Three Primary Colours	Name C.I. Disperse	Dyeing Depth (o.w.f %)	Dyeing Properties			Dyeing Fastness			
				T/P (HTHP)	TML (Thermosol)	Printing	Light (Xenon)	Sublimation (CH)	Washing (PES)	pH Value
100.		Grey TPP-HBL MIX	2.0	○	○	○	5	4-5	4-5	4-6
101.		Black TT-ECT MIX	4.0	○	◎	◎	6	4	4-5	4-8
102.		Black TT-XNT MIX	4.0	○	◎	◎	6	4	4-5	4-8
103.		Black TT-XSF MIX	4.0	○	△	◎	6	3-4	4-5	4-8
104.		Black TPP-2BL MIX	5.0	○	○	○	5-6	4-5	4-5	4-6
105.		Black TPP-4BL MIX	4.0	○	○	○	5-6	4-5	4-5	4-6
106.		Black TPP-GI MIX	4.0	○	○	○	5-6	4-5	4-5	4-6
107.		Black TT-XNR MIX	4.0	○	○	◎	6-7	4-5	4-5	4-8
108.		Black T-ECO MIX	4.0	○	◎	◎	6	3-4	4-5	4-8

Color Disclaimer: The colors shown are for reference only and may vary slightly from the actual product colors.

TIGA WARNA PRIMER

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